BRILLIANTS' CONVENT

Class -XII



Class XII

Physics

Activity

Prepare any one project out of the following given topics on A4 size sheet.

Investigatory Projects

1. To study various factors on which the internal resistance/EMF of a cell depends.
2. To study the variations in current flowing in a circuit containing an LDR because of a variation in (a) the power of the incandescent lamp, used to 'illuminate' the LDR (keeping all the lamps at a fixed distance). (b) the distance of a incandescent lamp (of fixed power) used to 'illuminate' the LDR.
3. To find the refractive indices of (a) water (b) oil (transparent) using a plane mirror, an equiconvex lens (made from a glass of known refractive index) and an adjustable object needle.
4. To investigate the relation between the ratio of (i) output and input voltage and (ii) number of turns in the secondary coil and primary coil of a self-designed transformer.
5. To investigate the dependence of the angle of deviation on the angle of incidence using a hollow prism filled one by one, with different transparent fluids.
6. To estimate the charge induced on each one of the two identical Styrofoam (or pith) balls suspended in a vertical plane by making use of Coulomb's law.
7. To study the factor on which the self-inductance of a coil depends by observing the effect of this coil, when put in series with a resistor/(bulb) in a circuit fed up by an A.C. source of adjustable frequency.
8. To study the earth's magnetic field using a compass needle -bar magnet by plotting magnetic field lines and tangent galvanometer.

Chemistry

An investigatory project has to be made on any topic of chemistry.

A few suggested Projects.

1. Study of the presence of oxalate ions in guava fruit at different stages of ripening.
2. Study the quantity of casein present in different samples of milk.
3. Preparation of soybean milk and its comparison with natural milk with respect to curd formation, the effect of temperature, etc.
4. Study of the effect of Potassium Bisulphate as a food preservative under various conditions (temperature, concentration, time, etc.)
5. Study of digestion of starch by salivary amylase and effect of pH and temperature on it.
6. Comparative study of the rate of fermentation of the following materials: wheat flour, gram flour, potato juice, carrot juice, etc.
7. Extraction of essential oils present in Saunf (aniseed), Ajwain (carum), Illaichi (cardamom).
8. Study of common food adulterants in fat, oil, butter, sugar, turmeric power, chilli powder and pepper.

Note: Any other investigatory project, which involves about 10 periods of work, can be chosen with the approval of the teacher.

Biology

Prepare a project on topic sustainable product of daily life on A4 size punch sheet with atleast 5 case studies having demographic and product details and it's application in society in the decorated manner.

Maths

Q1. A entrepreneur produces three products x, y, z which he sells in two markets. annual sales are indicated below:

Market		Products	
I	10,000	2,000	18,000
II	6,000	20,000	8,000

(a) If unit sale prices of x, y and z are Rs 2.50, Rs 1.50 and Rs 1.00, respectively, find the total revenue in each market with the help of matrix algebra.

(b) If the unit costs of the above three commodities are Rs 2.00, Rs 1.00 and 50 paise respectively. Find the gross profit.

Economics

Q1. Draw a profile of a successful entrepreneur.

Q2. Study & make report on a public sector undertaking and highlight its success/failure, by analyzing the factors responsible.

Accountancy

Q1. Briefly explain the ways by which you can arrange finance for start ups?

Q2. Compare the revenue and cost sheet of various start ups?

Business Studies

Q.1 Prepare a PowerPoint Presentation on the various start ups that have recently launched in India

Q.2 Make a word file on process on Entrepreneurship Process

English

Q1. Make notes on the chapter- The Last Lesson and The Lost Spring.

Q2. Write a notice informing students about the commencement of summer vacation of school, Give other necessary details.

Q3. Write the poem “Keeping Quiet” on A4 size sheet along with pictures.

Q4. Revise all the work done in the class.

Political Science

Q1. Read about all the related policies and laws in order to promote entrepreneurship skills amongst youths by the Government of India and prepare your report on it.

Q2. Prepare a PPT on how can entrepreneurship program can be promoted in India. Also what roles developed countries can play in it.

Computer Science

Q1. Do the following Python programs practically and make a single pdf file.

The format of the file is as follows:

- **Program heading (Question)**
- **Code of the program question**
- **Screenshot of the output**

SNo.	Name of Practical
1.	WAP to compute x n of given two integers x and n.
2.	WAP for calculating simple interest.
3.	WAP to accept a number from the user and display whether it is an even number or odd number.
4.	WAP to accept percentage of a student and display its grade accordingly.
5.	WAP to print Fibonacci series up to certain limit.
6.	WAP to display prime numbers up to a certain limit.
7.	P to accept a number, find and display whether it's a Armstrong number or not.
8.	WAP to accept a number and find out whether it is a perfect number or not.
9.	WAP to print the sum of the series $1+x^1/1!+x^2/2!+.....x^n/(n)!$- exponential series.
10.	WAP to print the following pattern: 1 1 2 1 2 3
11	WAP to accept a string and display whether it is a palindrome.

12.	WAP that counts the number of alphabets and digits, uppercase letters, lowercase letter, spaces and other characters in the string entered.
13.	WAP to accept a string (a sentence) and returns a string having first letter of each word in capital letter.
14.	WAP to remove all odd numbers from the given list.
15.	WAP to display second largest element of a given list.
16.	WAP to display cumulative elements of a given list.
17.	WAP to display frequencies of all the elements of a list.
18.	WAP in Python to display those strings which are string with 'A' of given list.
19.	WAP in Python to find and display the sum of all the values which are ending with 3 from a list.
20.	WAP to shift the negative number to left and the positive numbers to right.
21.	WAP to swap the content with next value divisible by 7.
22.	WAP to accept values from user and create a tuple.
23.	Write a program to input total number of sections and stream name in 11th class and display all information on the output screen.
24.	Write a Python program to input names of 'n' countries and their capital and currency, store it in a dictionary and display in tabular form. Also search and display for a particular country.

Informatics Practices

Q1. Do the following Python programs practically and make a single pdf file.

The format of the file is as follows:

- **Program heading (Question)**
- **Code of the program question**
- **Screenshot of the output**

SNo.	Name of the Practical
1	**Create a pandas series from a dictionary of values and an ndarray.**
2	**Given a Series, print all the elements that are above the 75th percentile.**
3	**Create a data frame for examination result and display row labels, column labels data types of each column and the dimensions**
4	**Filter out rows based on different criteria such as duplicate rows.**
5	**Locate the 3 largest values in a data frame.**
6	**Subtract the mean of a row from each element of the row in a Data Frame.**
7	**Replace all negative values in a data frame with a 0.**
8	**Importing and exporting data between pandas and CSV file.**
9	**Importing and exporting data between pandas and MySQL database.**
10	**Given the school result data, analyse the performance of the students on different parameters, e.g subject wise or class wise.**
11	**For the Data frames created above, analyze and plot appropriate charts with title and legend.**
12	**Program to plot frequency of marks using Line chart.**
13	**Program to draw two lines along with proper titles and legends.**

हिंदी

प्रश्न -1 प्रत्येक छात्र /छात्रा अपने अनुक्रमांक के अनुसार दिए गए विषयों से संबंधित एक परियोजना कार्य तैयार करेंगे, जिसमें 900 से 1000 शब्द होंगे। परियोजना कार्य बहुत ही स्वच्छ और त्रुटि रहित होना चाहिए।

1-अनुक्रमांक-1,2,3,4

कंप्यूटर एक जादुई पिटारा/सुविधाओं का भंडार: कंप्यूटर

2- अनुक्रमांक-5,6,7,8

बच्चों की शिक्षा में माता - पिता की भूमिका

3- अनुक्रमांक-9,10,11,12

जीना मुश्किल करती महंगाई

4- अनुक्रमांक -13,14,15,16

विश्व की शान बनता भारत

5- अनुक्रमांक- 17,18,19

मानव मन को सम्मोहित करते विज्ञापन/विज्ञापनों का पड़ता नई पीढ़ी पर प्रभाव

6- अनुक्रमांक- 20,21,22

बच्चों के जीवन में कोडिंग से होने वाले बदलाव

प्रश्न -2 पढ़ाए गए पाठ्यक्रम का पुनः अभ्यास करें।

Physical Education

Q1.Complete your Physical Education file as discussed in the class.

Q2.Revise the work done in the class so far.